

Linear Discriminant Analysis (LDA) – Theoretical MCQs & Answers

1. What is the primary objective of Linear Discriminant Analysis (LDA)?

- o A) Minimize reconstruction error
- o B) Maximize class separability
- o C) Minimize classification accuracy
- o D) Increase dimensionality

Answer: B) Maximize class separability

2. LDA is primarily used for:

- o A) Clustering
- o B) Regression
- o C) Dimensionality Reduction and Classification
- o D) Data Augmentation

Answer: C) Dimensionality Reduction and Classification

3. Which matrices are central to the formulation of LDA?

- o A) Covariance and Correlation Matrices
- o B) Within-Class Scatter and Between-Class Scatter Matrices
- o C) Adjacency and Incidence Matrices
- o D) Kernel and Gram Matrices

Answer: B) Within-Class Scatter and Between-Class Scatter Matrices

4. The within-class scatter matrix represents:

- o A) Separation between class means
- o B) Variance of samples within each class
- o C) Distance between clusters
- o D) Prediction error

Answer: B) Variance of samples within each class

5. The between-class scatter matrix measures:

- o A) Variability within classes
- o B) Separation among class means
- o C) Noise variance
- o D) Reconstruction loss

Answer: B) Separation among class means

6. LDA seeks projection directions that:

- o A) Minimize both within-class and between-class variance

- o B) Maximize within-class variance
- o C) Maximize between-class variance while minimizing within-class variance
- o D) Ignore class labels

Answer: C) Maximize between-class variance while minimizing within-class variance

7. For a dataset with C classes, the maximum number of LDA components is:

- o A) C
- o B) C + 1
- o C) C – 1
- o D) Equal to the number of features

Answer: C) C – 1

8. Which of the following is an assumption of classical LDA?

- o A) Classes have identical covariance matrices
- o B) Classes have different covariance matrices
- o C) Features are independent and binary
- o D) Data must be non-linear

Answer: A) Classes have identical covariance matrices

9. LDA is a supervised technique because:

- o A) It uses unlabeled data
- o B) It requires class labels
- o C) It uses reinforcement signals
- o D) It performs clustering

Answer: B) It requires class labels

10. In LDA, eigenvectors are obtained from:

- o A) Covariance Matrix
- o B) Correlation Matrix
- o C) $S_W^{-1}S_B$
- o D) Gram Matrix

Answer: C) $S_W^{-1}S_B$

11. Compared to PCA, LDA:

- o A) Does not reduce dimensions
- o B) Ignores class labels
- o C) Uses class information
- o D) Requires more dimensions

Answer: C) Uses class information

12. Which criterion is maximized in Fisher's Linear Discriminant?

- o A) Reconstruction Error
- o B) Variance Ratio

- o C) Between-Class Scatter to Within-Class Scatter Ratio
- o D) Mean Squared Error

Answer: C) Between-Class Scatter to Within-Class Scatter Ratio

13. LDA assumes data from each class follows:

- o A) Uniform Distribution
- o B) Poisson Distribution
- o C) Gaussian Distribution
- o D) Exponential Distribution

Answer: C) Gaussian Distribution

14. If there are two classes, the maximum number of discriminant components is:

- o A) 0
- o B) 1
- o C) 2
- o D) Number of features

Answer: B) 1

15. Which statement best differentiates PCA and LDA?

- o A) PCA is supervised; LDA is unsupervised
- o B) PCA maximizes variance; LDA maximizes class separability
- o C) PCA requires labels; LDA does not
- o D) Both optimize the same objective

Answer: B) PCA maximizes variance; LDA maximizes class separability

16. What happens if class labels are randomly shuffled before applying LDA?

- o A) Performance generally degrades
- o B) Performance improves
- o C) No effect
- o D) Number of features increases

Answer: A) Performance generally degrades

17. Which matrix must generally be invertible in standard LDA?

- o A) Between-Class Scatter Matrix
- o B) Within-Class Scatter Matrix
- o C) Correlation Matrix
- o D) Distance Matrix

Answer: B) Within-Class Scatter Matrix

18. LDA projects data onto:

- o A) Directions of maximum variance
- o B) Random directions
- o C) Directions maximizing discrimination among classes

- o D) Kernel feature space only

Answer: C) Directions maximizing discrimination among classes

19. In pattern recognition, LDA is often applied:

- o A) Before classification
- o B) After classification
- o C) Only during testing
- o D) Only for clustering

Answer: A) Before classification

20. Which of the following can be a limitation of LDA?

- o A) Requires labeled data
- o B) Assumes linear boundaries
- o C) Sensitive to covariance estimation
- o D) All of the above

Answer: D) All of the above

21. LDA can be viewed as:

- o A) Feature Extraction Technique
- o B) Data Compression Technique
- o C) Classification Technique
- o D) All of the above

Answer: D) All of the above

22. The rank of the between-class scatter matrix is at most:

- o A) Number of Features
- o B) Number of Samples
- o C) Number of Classes Minus One
- o D) Twice the Number of Classes

Answer: C) Number of Classes Minus One

23. Which application commonly uses LDA?

- o A) Face Recognition
- o B) Speech Recognition
- o C) Medical Diagnosis
- o D) All of the above

Answer: D) All of the above

24. In Fisher's criterion, a good projection should:

- o A) Bring class means closer together
- o B) Increase overlap among classes
- o C) Separate class means and reduce class spread
- o D) Ignore variance

Answer: C) Separate class means and reduce class spread

25. LDA is most effective when:

- ☐ A) Classes are well separated
- ☐ B) Data is completely random
- ☐ C) No labels are available
- ☐ D) Features are constant

Answer: A) Classes are well separated

26. Which of the following affects LDA performance?

- ☐ A) Class imbalance
- ☐ B) Outliers
- ☐ C) Small sample size
- ☐ D) All of the above

Answer: D) All of the above

27. The generalized eigenvalue problem in LDA is solved to obtain:

- ☐ A) Cluster Centroids
- ☐ B) Principal Components
- ☐ C) Discriminant Directions
- ☐ D) Decision Trees

Answer: C) Discriminant Directions

28. LDA is likely to perform poorly when:

- ☐ A) Classes are linearly separable
- ☐ B) Covariance assumptions are heavily violated
- ☐ C) Labels are accurate
- ☐ D) Features are informative

Answer: B) Covariance assumptions are heavily violated

29. Which quantity is minimized in the denominator of Fisher's criterion?

- ☐ A) Between-Class Scatter
- ☐ B) Within-Class Scatter
- ☐ C) Total Variance
- ☐ D) Correlation

Answer: B) Within-Class Scatter

30. In multiclass classification, LDA creates:

- ☐ A) One discriminant axis only
- ☐ B) Multiple discriminant axes up to $C - 1$
- ☐ C) One axis per feature
- ☐ D) No projection axes

Answer: B) Multiple discriminant axes up to $C - 1$

31. LDA is particularly useful when:

- ☐ A) Number of features is very small
- ☐ B) Feature reduction is needed before classification
- ☐ C) Data contains no labels
- ☐ D) Only clustering is required

Answer: B) Feature reduction is needed before classification

32. Which assumption is shared by many LDA formulations?

- ☐ A) Equal class priors
- ☐ B) Equal covariance matrices across classes
- ☐ C) Independent samples only
- ☐ D) Nonlinear decision surfaces

Answer: B) Equal covariance matrices across classes

33. The output of LDA is typically:

- ☐ A) Cluster Assignments
- ☐ B) Reduced-Dimensional Feature Vectors
- ☐ C) Regression Coefficients
- ☐ D) Association Rules

Answer: B) Reduced-Dimensional Feature Vectors

34. Why is LDA often preferred over PCA for classification tasks?

- ☐ A) It ignores variance
- ☐ B) It uses label information
- ☐ C) It requires fewer samples
- ☐ D) It is unsupervised

Answer: B) It uses label information

35. If there are 5 classes in a dataset, the maximum number of discriminant vectors produced by LDA is:

- ☐ A) 3
- ☐ B) 4
- ☐ C) 5
- ☐ D) 6

Answer: B) 4

36. Which type of learning does LDA belong to?

- ☐ A) Supervised Learning
- ☐ B) Unsupervised Learning
- ☐ C) Reinforcement Learning
- ☐ D) Self-Supervised Learning

Answer: A) Supervised Learning

37. What role do class means play in LDA?

- ☐ A) No role
- ☐ B) Used to compute between-class scatter
- ☐ C) Used only for normalization
- ☐ D) Used only during testing

Answer: B) Used to compute between-class scatter

38. Which of the following is true about LDA decision boundaries?

- ☐ A) Always nonlinear
- ☐ B) Typically linear
- ☐ C) Always quadratic
- ☐ D) Always circular

Answer: B) Typically linear

39. LDA can suffer from the "small sample size" problem when:

- ☐ A) Number of samples is much smaller than number of features
- ☐ B) Number of features is zero
- ☐ C) Classes are balanced
- ☐ D) Data is normalized

Answer: A) Number of samples is much smaller than number of features

40. Which statement about LDA is correct?

- ☐ A) It ignores class membership
- ☐ B) It maximizes variance without labels
- ☐ C) It seeks projections that improve class discrimination
- ☐ D) It is purely a clustering algorithm

Answer: C) It seeks projections that improve class discrimination